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SLIATE

SRI LANKA INSTITUTE OF ADVANCED TECHNOLOGICAL EDUCATION

(Established in the Ministry of Higher Education, vide in Act No. 29 of 1995)

Higher National Diploma in Information Technology

Second Year, First Semester Examination – 2023

HNDIT3072 – Statistics for IT

Instructions for Candidates:

Answer any five (05) questions.

Every question carries 20 marks

You are allowed to use non-programable calculator.

No. of questions : 06

No. of pages : 05

Time : Three (03) hours

ANSWER SCRIPT

Question 01.

(i). Define the following terms;

(a). Experiment

(b). Event

(c). Mutually exclusive events

(d). Sample space [04 Marks]

(a). Experiment:
An experiment is a process or action that results in one or more outcomes, and its result is subject to uncertainty. In probability theory, an experiment is typically conducted to observe outcomes and analyze their likelihood. For example, tossing a coin or rolling a die.

(b). Event:
An event is a specific outcome or a set of outcomes from an experiment. It is a subset of the sample space. For example, when rolling a die, an event could be rolling an even number or rolling a number greater than 4.

(c). Mutually exclusive events:
Mutually exclusive events are events that cannot occur at the same time. If one event occurs, the other cannot. For example, when flipping a coin, getting "heads" and getting "tails" are mutually exclusive events because both cannot happen simultaneously.

(d). Sample space:
The sample space is the set of all possible outcomes of an experiment. For example, when tossing a fair coin, the sample space is {heads, tails}, as these are the only two possible outcomes.

(01 marks * 4 = 04 marks)

(ii). A bag contains 5 red balls, 3 blue balls and 2 green balls.

(a). What is the sample space of this experiment? [02 Mark]

(b). If a ball is picked randomly from the bag, what are the probabilities of selecting each type of balls. (red, blue, green)? [02 Marks]

(c). What is the probability of getting either a red ball or green ball? [02 Marks]

(d). If two balls are picked randomly from the bag without replacement, what is the probability of selecting first red ball and second blue ball? [02 marks]

(a). Sample Space (S) = {Red, Red, Red, Red, Red, Blue, Blue, Blue, Green, Green}

[02 Marks]

(b).

$$\textbf{P (red)} = \frac{number\ of\ red\ balls}{total\ number\ of\ balls} = \frac{5}{10} = \frac{1}{2}$$

$$\textbf{P (blue)} = \frac{number\ of\ blue\ balls}{total\ number\ of\ balls} = \frac{3}{10}$$

$$\textbf{P (green)} = \frac{number\ of\ green\ balls}{total\ number\ of\ balls} = \frac{2}{10} = \frac{1}{5}$$

[02 Marks]

(c).

$$\textbf{P(red or green)} = \textbf{P(red) + P(green)}$$

$$= \frac{5}{10} + \frac{2}{10} = \frac{7}{10}$$

[02 Marks]

(d).

$$\textbf{P (red first)} = \frac{5}{10}$$

$$\textbf{P (blue second)} = \frac{3}{9}$$

$$\textbf{P (red first and blue second)} = \frac{5}{10} * \frac{3}{9}$$

$$= \frac{15}{90} = \frac{1}{6}$$

[02 Marks]

(iii). In a group of 150 employees at a company, the probability of selecting a male employee is 0.6. Additionally, 45 employees (both male and female) have received a leadership award.

(a). What is the probability of selecting a female employee? Show your work using a probability rule. [02 Marks]

(b). What is the probability of selecting an employee who has not received a leadership award? [02 Marks]

(c).). If the probability of selecting a male employee who received a leadership award is 0.18, find the probability of the selected employee having received a leadership award given that they are male. [04 marks]

(a). P(female) = 1 – P(male)

= 1- 0.6

= 0.4 [02 marks]

(b). Employees without leadership award=150−45=105

P(not leadership award) = $\frac{\text{Employees without leadership award}}{\boldsymbol{Total\ employees}}$

$= \frac{\mathbf{105}}{\mathbf{150}} = \mathbf{0.7}$ [02 marks]

(c). P(leadership award | male) = $\frac{\text{P(male and leadership award}}{\boldsymbol{P(male)}}$

$= \frac{\mathbf{0.18}}{\mathbf{0.6}} = \mathbf{0.3}$ [02 marks]

[20 Marks]

Question 02.

(i). Define a Probability Distribution? [02 Marks]

A **Probability Distribution** describes how the probabilities are distributed over the possible values of a random variable. It is a mathematical function that gives the probability of each possible outcome of an experiment.

The sum of all probabilities must equal 1: $\sum \boldsymbol{P(X)} = \mathbf{1}$

Each individual probability must lie between 0 and 1: $\mathbf{0 \leq P(X) \leq 1}$

[02 Marks]

(ii). Differentiate between Binomial Probability Distribution and Poison Probability Distribution. [02 Marks]

Aspect	Binomial Probability Distribution	Poisson Probability Distribution
Definition	Describes the probability of a fixed number of successes in a given number of independent trials.	Describes the probability of a given number of events occurring in a fixed interval of time or space.
Key Parameters	n: Number of trials p: Probability of success per trial	λ: Average number of occurrences in the interval (mean)
Random Variable	Discrete, counting the number of successes in n trials	Discrete, counting the number of events in a fixed interval
Conditions	- Trials are independent - Only two outcomes: success or failure - Probability of success remains constant	- Events occur independently - Events happen at a constant average rate - No two events occur simultaneously
Formula for Probability	$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$	$P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$
Mean (μ)	$\mu = np$	$\mu = \lambda$
Variance (σ^2)	$\sigma^2 = np(1-p)$	$\sigma^2 = \lambda$
Shape of Distribution	Symmetrical when $p = 0.5$; skewed for smaller or larger p	Right-skewed, especially when λ is small
Application	Used when trials are fixed, and the probability of success is constant (e.g., flipping coins, defect rates).	Used for counting rare events over a continuous interval (e.g., phone calls, accidents).

[02 marks for any proper explanation]

(iii). A company manufactures light bulbs, and the probability distribution for the number of defective light bulbs in a batch of 5 bulbs is given below.

No. of defective bulbs (x)	0	1	2	3	4	5
P(x)	0.2	0.3	0.25	a	0.1	0.05

Find;

(a). the value a. [04 Marks]

P(0) + P(1) + + P(2) + P(3) + P(4) + P(5) = 1

0.2 + 0.3 + 0.25 + a + 0.1 + 0.05 =1 (02 marks)

a = 0.1 (02 marks)

(b). Calculate the expected number of defective bulbs in a batch of 5 bulbs.[04 Marks]

E(X) = $\sum x$. P (x) (01 marks)

= (0*0.2) + (1* 0.3) + (2*0.25) + (3*0.1) + (4*0.1) + (5*0.05) (02 marks)

= 1.75 (01 marks)

(iv). A football player has a 70% chance of making a free throw. Suppose the player takes 6 free throws. Using the **binomial probability distribution**, find:

(a). the probability of making exactly 4 free throws. [04 Marks]

P(X=x) $= \binom{n}{x} P^x q^{n-x}$

P(X=x) $= \binom{6}{4} 0.7^4 \, 0.3^{6-4}$ (02 marks)

=0.324135 (02 marks)

(b). The expected value and variance of the number of successful free throws.

[04 Marks]

$E(x) = np$
$= 6 * 0.7$
$= 4.2$ (02 marks)

(c). List 02 key attributes of Binomial probability distribution. [02 Marks]

- Fixed Number of Trials
- Two Possible Outcomes per Trial
- Constant Probability of Success
- Independence of Trials

(01 mark * 02 points = 02 marks)

[20 Marks]

Question 03.

(i). Differentiate between Primary data and Secondary data. [04 Marks]

Aspect	Primary Data	Secondary Data
Definition	Data collected firsthand by the researcher for a specific purpose.	Data that has already been collected, processed, and published by someone else.
Source	Original source (e.g., surveys, experiments, interviews).	Derived from existing sources (e.g., books, reports, websites).
Collection Process	Requires direct effort to gather raw data.	Involves using already available data.
Cost and Time	Generally more time-consuming and expensive to collect.	Less expensive and time-saving since the data already exists.
Accuracy and Reliability	Usually more accurate and reliable as it is specific to the purpose.	May be less reliable or relevant, as it was collected for a different purpose.
Examples	Surveys, experiments, direct observations, focus groups.	Research articles, government reports, books, databases.

[04 marks for any proper explanation including at least 02 key points]

(ii). Explain the importance of samplingexplain 02 methods of probabilistic sampling.

[04 Marks]

Importance

- **Cost-Effectiveness:** Sampling reduces the cost of research by analyzing only a portion of the population instead of the entire group.
- **Time Efficiency:** Studying a sample is quicker and more practical than studying the entire population.
- **Feasibility:** In many cases, it is impossible or impractical to collect data from the entire population.
- **Accuracy:** Properly selected samples can provide highly accurate results and reliable conclusions.
- **Focus:** Enables researchers to focus on key characteristics of interest within the population.
- **Resource Optimization:** Sampling makes efficient use of available resources such as manpower and tools.

methods of probabilistic sampling:

- **Simple Random Sampling** : Each individual in the population has an equal and independent chance of being selected.
- **Systematic Sampling**: Individuals are selected at regular intervals from a list of the population.
- **Stratified Sampling**: The population is divided into subgroups (strata) based on shared characteristics, and random samples are taken from each stratum.
- **Cluster Sampling**: The population is divided into clusters (often geographically), and a random sample of clusters is selected. All individuals within the selected clusters are studied.

[02 marks for explanation of importance + 02 marks for explanation of 02 methods]

(iii). Sample of 25 employees shows their monthly salaries (in Rs. 000):

35	40	50	45	30	60	65	55	70	35
80	85	70	40	50	45	90	100	95	50
55	60	35	40	45					

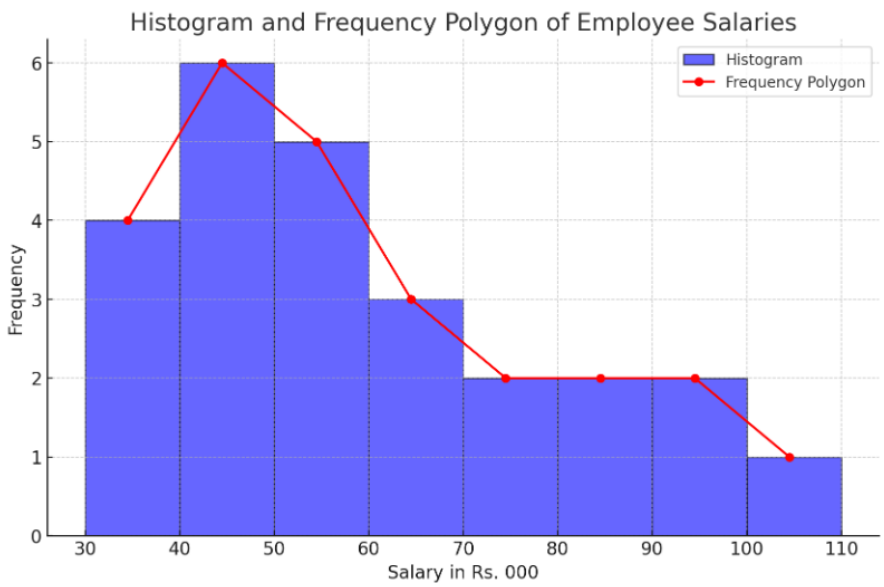
(a). Construct a frequency distribution with a class width of 10, starting at 30.

[04 Marks]

Class Interval (Salary in Rs. 000)	Tally mark	Frequency
30-39		4
40-49		6
50-59		5
60-69		4
70-79		3
80-89		2
90-99		2
100-109		1

(0.5 Marks for each record of the frequency table * 8 = 04 marks)

(b). Construct the Histogram and the Frequency Polygon. [04 Marks]



(02 marks for Histogram + 02 marks for frequency polygon = 04 marks)

(iv). Provide short answers for followings.

(a). How do you identify the modal class from a histogram? [02 Marks]

the class with the highest bar represents the modal class (02 marks)

(b). What does a *Histogram* represent? [02 Marks]

a type of bar chart that represents the **frequency distribution** of a dataset. (02 marks)

[20 Marks]

Question 04.

(i).

(a). Define the Mean and Mode of a dataset. [02 Marks]

Mean of a Dataset:

The **mean** (or **arithmetic mean**) is the average of all the values in a dataset. It is calculated by summing up all the values and dividing by the total number of values in the dataset.

(01 marks)

Mode of a Dataset:

The **mode** is the value that appears most frequently in a dataset. (01 marks)

(b). Explain the importance of central tendency measurements in statistics. [02 Marks]

- provides a simple summary of the data by representing a single, central value. This helps to simplify complex data, making it easier to understand and compare.
- By calculating the central tendency for different datasets or groups, you can make comparisons.
- Central tendency helps in making informed decisions.
- The measurements of central tendency reveal whether a dataset tends to cluster around a particular value.
- simplifies large datasets by focusing on the representative value.

(02 marks for any key poin)

(ii). The number of books read by students during their vacation are as follows;

10, 12, 5, 2, 82, 10, 8, 4, 5, 6

(a). Calculate the average no of books read by a student. [01 Mark]

(b). Calculate the median value the dataset. [01 Marks]

(c). What is/are the mode value(s) of above dataset? [01 Marks]

(d). Briefly explain the effect does value 82 in the above dataset have on the average value. [01 Marks]

(a). Mean = $\frac{144}{10} = 14.4$ (01 marks)

(b). **Sorted dataset**: **2, 4, 5, 5, 6, 8, 10, 10, 12, 82**

Median position: $\frac{(10+1)}{2} = 5.5^{th}$ position

Median = $\frac{6+8}{2} = 7$ (01 marks)

(c). **mode** is **5 and 10**. (01 marks)

(d). The value **82** is an **outlier** in this dataset because it is much higher than the other values.

Effect on the mean: The presence of 82 in the dataset **increases the average (mean)** significantly. Without this outlier, the mean would have been much lower.

The mean might not accurately represent the typical number of books read by a student due to the influence of this outlier. (01 marks)

(iii). Number of hours worked by 100 workers per month are as follows;

Worked hours	No. of workers
30-40	5
40-50	15
50-60	20
60-70	30
70-80	20

80-90	10

(a). Calculate the average number of hours. Worked by a worker. [04 Marks]

Worked Hours	No. of Workers	Midpoint (x)	Frequency (f)	f * x
30-40	5	35	5	175
40-50	15	45	15	675
50-60	20	55	20	1100
60-70	30	65	30	1950
70-80	20	75	20	1500
80-90	10	85	10	850

- Total frequency (Σf) = 5 + 15 + 20 + 30 + 20 + 10 = **100**
- Sum of f * x (Σf * x) = 175 + 675 + 1100 + 1950 + 1500 + 850 = **6250**

Mean = $\frac{6250}{100}$ = 62.5 hrs

(01 marks for tabular calculation + 01 mark for equation +02 mark for correct answer = 04 marks)

(b). Calculate the Median value of worked hours. [04 Marks]

Worked Hours	No. of Workers	Cumulative Frequency
30-40	5	5
40-50	15	20
50-60	20	40
60-70	30	70
70-80	20	90
80-90	10	100

Median position: 100/2 = 50

median class is **60-70**.

$$\text{Median} = L + \left(\frac{\frac{n}{2} - CF}{f}\right) * C$$

$$= 60 + \left(\frac{\frac{100}{2} - 40}{30}\right) * 10 \qquad = 63.33 \text{ hours}$$

(01 marks for cumulative frequency + 01 mark for equation +02 mark for correct answer = 04)

(c). Calculate the most frequent number of hours (mode) worked by workers.

[04 Marks]

The highest frequency is **30**, which corresponds to the class **60-70.**

Hence the **mode class** is **60-70** hours.

$$\text{Mode} = L + \left(\frac{Da}{Db + Da}\right) * C$$

$$= 60 + \left(\frac{(30-20)}{(30-20)+930-20)}\right) * 10$$

$$= 60 + \left(\frac{10}{20}\right) * 10$$

$$= 65$$

(01 marks for identifying model class + 01 mark for equation +02 mark for correct answer = 04) marks)

[20 Marks]

Question 05.

(i).

(a). Define the *Variance* and Standard Deviation. [02 Marks]

Variance is a statistical measure of the spread or dispersion of a set of data points. It is the average of the squared differences from the mean. (01 marks)

Standard deviation is the square root of the variance and provides a measure of the average distance of data points from the mean, but in the original units of the data. (01 marks)

(b). Why are the variance and SD important in data analysis? [02 Marks]

- **Variance** and **standard deviation** give insights into the **variability** or **spread** of the data.
- They are critical in assessing **risk**, **predictability**, and **data consistency**.
- These measures are used to **compare datasets**, **identify outliers**, and make informed **decisions** in various fields.

- They play a crucial role in **statistical modeling** and **hypothesis testing**.

(02 marks for any proper explanation)

(ii). "Mean is always a good statistic measure in every statistical analysis". Do you agree with the statement? Justify your answer. [04 Marks]

I **do not fully agree** with the statement that "Mean is always a good statistic measure in every statistical analysis."

The **mean** is highly sensitive to extreme values (outliers). Even one or a few extreme values can significantly skew the mean, making it unrepresentative of the dataset as a whole. In such cases, the **median** or **mode** may provide more accurate and meaningful insights.

(01 marks for disagreement with initiated explanation + 03 marks for correct justification)

(iii). The ages of few of staff in a company are as follows;

25 30 35 40 45 50 55 60

(a). Calculate the Inter quartile range. [04 Marks]

(b). Calculate the Variance of the ages of staff. [04 Marks]

(c). Calculate the Standard Deviation of the ages of staff. [02 Marks]

(d). Calculate the Co-efficient of Variance. [02 Marks]

(a). 25, 30, 35, 40, 45, 50, 55, 60

Q1th position = $\frac{n+1}{4}$ = = $\frac{9}{4}$ = 2.25th position =2.5th position

Hence Q1 = $\frac{30+35}{2}$ = 32.5

Q2th position = $\frac{(n+1)*3}{4}$ = = $\frac{9}{4}$ = 6.75th position= 6.5th position

Hence Q2 = $\frac{50+55}{2}$ = 52.5

ICR = Q3 -Q1 = 52.5 – 32.5 = 20

(01 mark for Q1 + 01 mark for Q3 + 02 marks for ICR = 04 marks)

[20 Marks]

Question 06.

(i).

(a). Define correlation. [02 Marks]

a statistical measure that describes the degree and direction of the relationship between two or more variables. (02 marks)

(b). Differentiate between positive correlation and negative correlation. [02 Marks]

Aspect	Positive Correlation	Negative Correlation
Direction	Both variables increase or decrease together	One variable increases, the other decreases
Correlation Coefficient	Between 0 and +1	Between 0 and -1
Graphical Trend	Upward sloping (left to right)	Downward sloping (left to right)
Example	Height and weight, study time and grades	Time spent on social media and productivity

(02 marks)

(c). Provide an example of a Dependent variable and Independent variable within a selected context. [02 Marks]

(ii). The following data shows the number of hours studied (X) and marks obtained (Y) by students:

Hours Studied (x)	Marks Obtained (y)
2	30
3	40
4	50
5	70
6	80

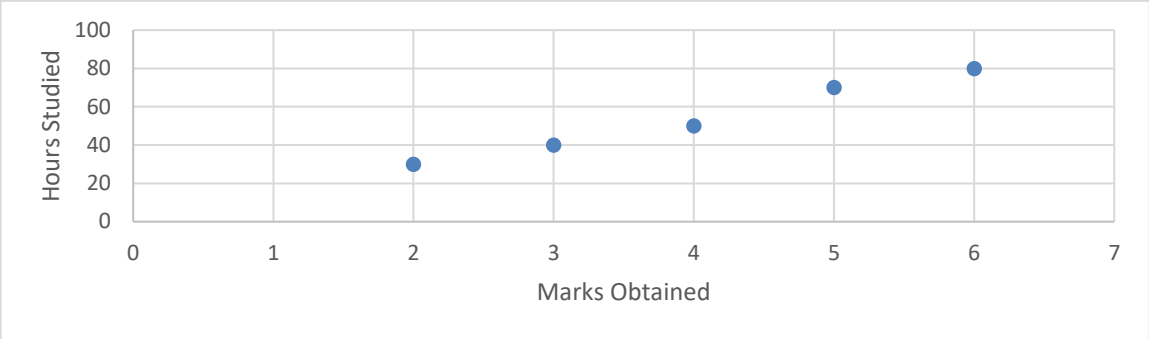
Considering the provided information;

(a). Draw a scatter plot and identify the type of correlation. [04 Marks]

(b). Calculate the Correlation Coefficient. (Use Spearmen's Rank Correlation Coefficient method) explain the correlation between x and y. [05 Marks]

(c). Find the Regression equation Y= a+ bX. [05 Marks]

(a).



(02 marks)

positive correlation (02 marks)

(b).

X (Hours)	Y (Marks)	Rank of X	Rank of Y	$d = \text{Rank of X} - \text{Rank of Y}$	d^2
2	30	1	1	0	0
3	40	2	2	0	0
4	50	3	3	0	0
5	70	4	4	0	0
6	80	5	5	0	0

(02 marks)

$r_s = 1 - \frac{6\sum d^2}{n(n^2-1)} = 1 - \frac{6\sum 0}{5(5^2-1)} = 1-0 = 1$ (02 marks)

The correlation is **1**, meaning that the relationship between X (hours studied) and Y (marks obtained) is **perfectly positive**. (01 marks)

(c).

$$\sum X = 2 + 3 + 4 + 5 + 6 = 20$$

$$\sum Y = 30 + 40 + 50 + 70 + 80 = 270$$

$$\sum XY = 2 \times 30 + 3 \times 40 + 4 \times 50 + 5 \times 70 + 6 \times 80 = 1210$$

$$\sum X^2 = 2^2 + 3^2 + 4^2 + 5^2 + 6^2 = 90$$

$n = 5$ (number of data points)

$\Sigma\, y = na + b\Sigma x$

$\Sigma\, xy = a\Sigma\, x + b\Sigma\, x^2$

270 = 5a + 20b (01 mark)

1210 = 20a + 90b (01 mark)

a = 2, b = 13 (02 mark)

Y = a + bx

Y= 2 + 13 x (01 mark)

[20 Marks]